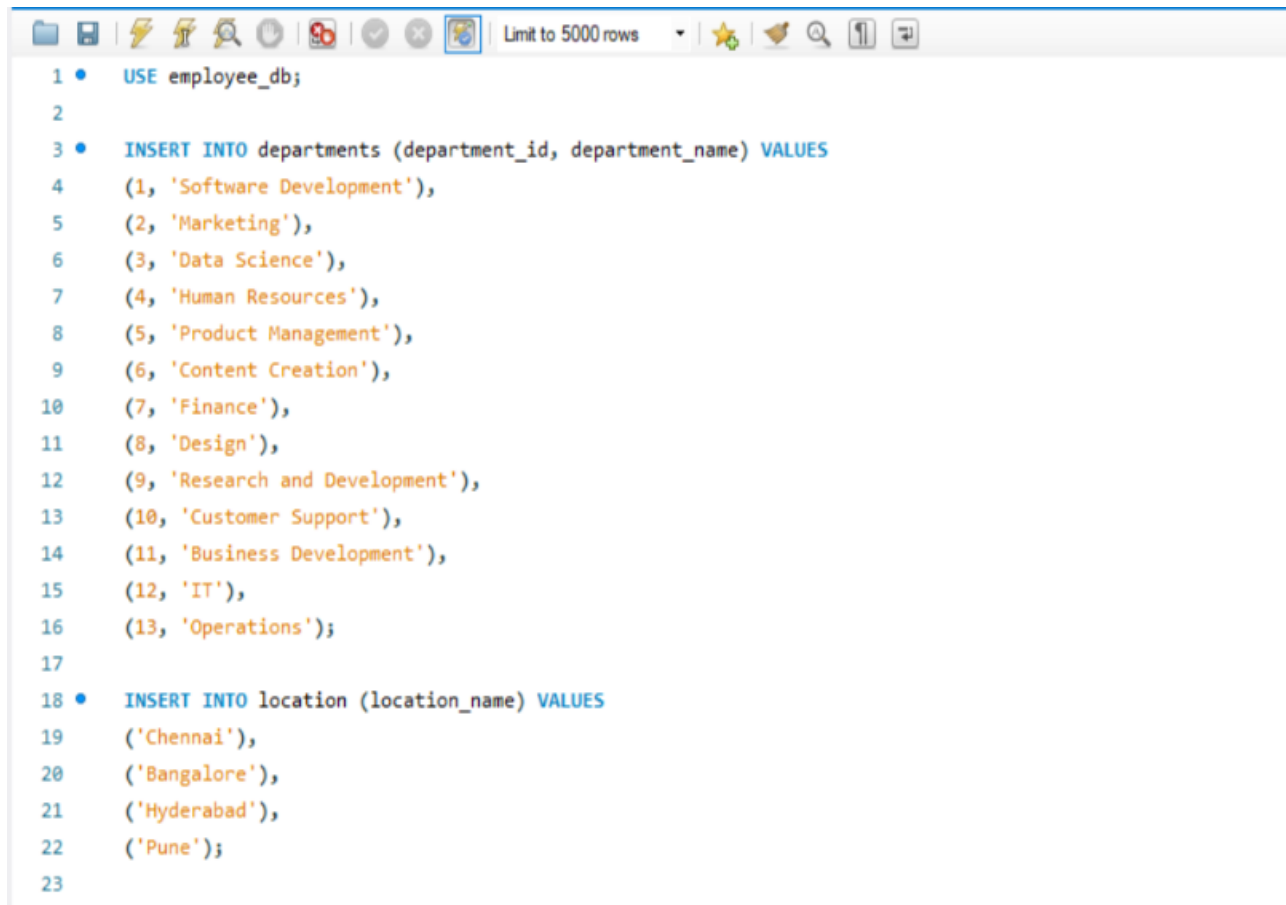


MY SQL ASSIGNMENT-2



The image shows a screenshot of a SQL IDE window. The title bar includes a toolbar with icons for file operations, execution, and search, along with a text input field that says "Limit to 5000 rows". The main area contains SQL code with line numbers on the left. The code consists of two SQL statements: a 'USE' statement and two 'INSERT INTO' statements. The first 'INSERT INTO' statement is for the 'departments' table, listing 13 departments with their IDs and names. The second 'INSERT INTO' statement is for the 'location' table, listing four locations: Chennai, Bangalore, Hyderabad, and Pune.

```
1 • USE employee_db;
2
3 • INSERT INTO departments (department_id, department_name) VALUES
4     (1, 'Software Development'),
5     (2, 'Marketing'),
6     (3, 'Data Science'),
7     (4, 'Human Resources'),
8     (5, 'Product Management'),
9     (6, 'Content Creation'),
10    (7, 'Finance'),
11    (8, 'Design'),
12    (9, 'Research and Development'),
13    (10, 'Customer Support'),
14    (11, 'Business Development'),
15    (12, 'IT'),
16    (13, 'Operations');
17
18 • INSERT INTO location (location_name) VALUES
19     ('Chennai'),
20     ('Bangalore'),
21     ('Hyderabad'),
22     ('Pune');
23
```

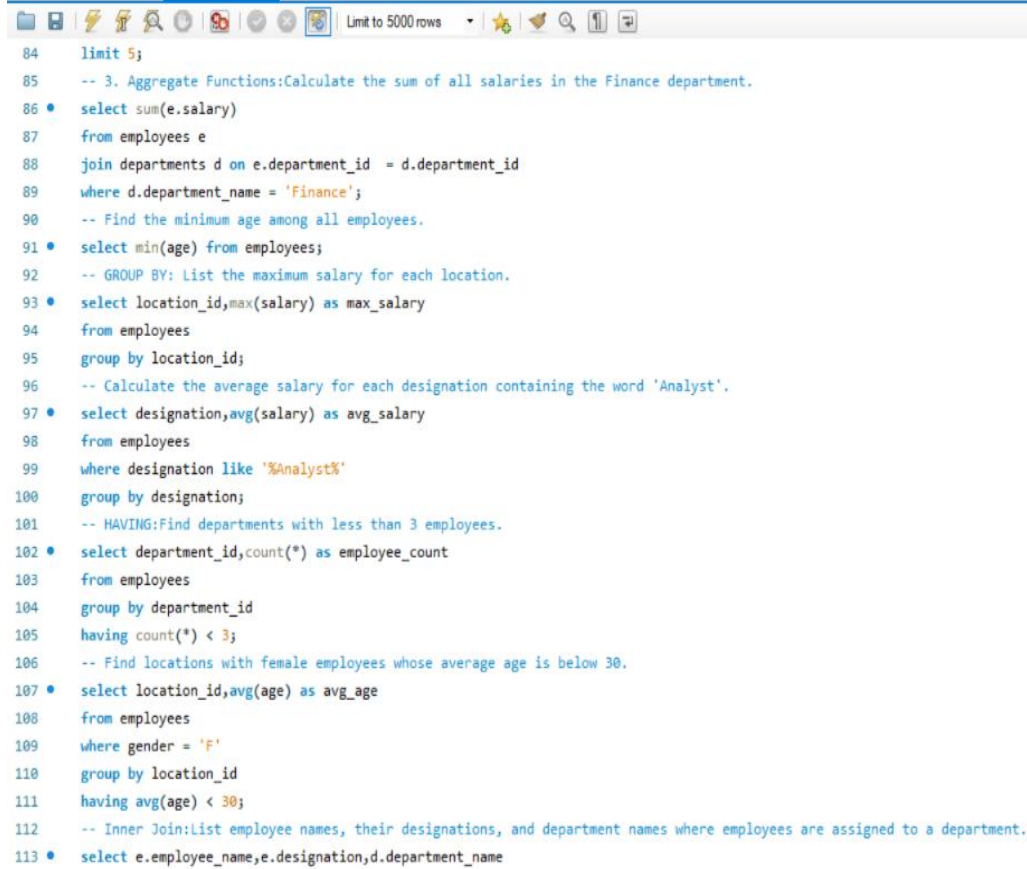
Limit to 5000 rows

```
24 • INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary) VALUES
25 (5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
26 (5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
27 (5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
28 (5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
29 (5005, 'Anvi Chaudhary', 'F', 25, '2015-09-11', 'Business Development Executive', 11, 1, 55000),
30 (5006, 'Dhruv Shetty', 'M', 28, '2015-11-20', 'UI Developer', 8, 2, 65000),
31 (5007, 'Anushka Singh', 'F', 32, '2016-01-15', 'Marketing Manager', 2, 3, 90000),
32 (5008, 'Diya Jha', 'F', 27, '2016-03-05', 'Graphic Designer', 8, 4, 70000),
33 (5009, 'Kiaan Desai', 'M', 30, '2016-05-20', 'Sales Executive', 11, 3, 55000),
34 (5010, 'Atharv Yadav', 'M', 29, '2016-07-10', 'Systems Administrator', 12, 4, 80000),
35 (5011, 'Saarvi Patel', 'F', 28, '2016-09-20', 'Marketing Analyst', 2, 1, 60000),
36 (5012, 'Myra Verma', 'F', 26, '2016-11-05', 'Operations Manager', 13, 2, 95000),
37 (5013, 'Arnav Rao', 'M', 33, '2017-01-20', 'Customer Success Manager', 10, 3, 75000),
38 (5014, 'Vihaan Mohan', 'M', 30, '2017-03-10', 'Supply Chain Analyst', 10, 2, 60000),
39 (5015, 'Ishaan Kumar', 'M', 27, '2017-05-20', 'Financial Analyst', 7, 1, 85000),
40 (5016, 'Zoya Khan', 'F', 31, '2017-07-05', 'Legal Counsel', 4, 4, 100000),
41 (5017, 'Kabir Nair', 'M', 28, '2017-09-11', 'IT Support Specialist', 12, 2, 80000),
42 (5018, 'Ishan Mishra', 'M', 25, '2017-11-20', 'Research Scientist', 9, 3, 75000),
43 (5019, 'Ishika Patel', 'F', 29, '2018-01-15', 'Talent Acquisition Specialist', 4, 4, 55000),
44 (5020, 'Aarav Nair', 'M', 32, '2018-03-05', 'Software Engineer', 1, 1, 90000),
45 (5021, 'Advik Kapoor', 'M', 26, '2018-05-20', 'Finance Analyst', 7, 3, 85000),
46 (5022, 'Aadhya Iyengar', 'F', 28, '2018-07-10', 'HR Specialist', 4, 4, 60000),
47 (5023, 'Anika Paul', 'F', 30, '2018-09-20', 'Public Relations Specialist', 2, 2, 70000),
48 (5024, 'Aryan Shetty', 'M', 27, '2018-11-05', 'Product Manager', 5, 1, 95000),
49 (5025, 'Avni Iyengar', 'F', 31, '2019-01-20', 'Data Scientist', 3, 4, 100000),
50 (5026, 'Vivaan Singh', 'M', 29, '2019-03-10', 'Business Analyst', 3, 2, 75000),
51 (5027, 'Ananya Paul', 'F', 32, '2019-05-20', 'Content Writer', 6, 3, 60000),
52 (5028, 'Anaya Kapoor', 'F', 26, '2019-07-05', 'Event Coordinator', 6, 1, 60000),
53 (5029, 'Arjun Kumar', 'M', 33, '2019-09-11', 'Quality Assurance Analyst', 12, 2, 80000),
```

```

54  (5030, 'Sara Iyer', 'F', 28, '2019-11-20', 'Project Manager', 5, 1, 90000);
55  -- distinct values
56  -- a query to retrieve distinct salaries from the Employees table.
57  • select distinct salary
58  from employees;
59  -- Alias (AS):
60  -- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively
61  • select age as Employee_age from employees;
62  • select salary as Employee_Salary from employees;
63  -- Where Clause & Operators: Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
64  • select salary from employees
65  where salary > 50000 and hire_date < '2016-01-01';
66  -- Find the employee whose designation is missing and fill it with "DataScientist".
67  • select*from employees
68  where designation is null;
69  -- updated employees table
70  • set sql_safe_updates = 0;
71  • update employees
72  set designation = 'Data Scientist'
73  where designation is null;
74  -- order by
75  -- Find employees sorted by department ID in ascending order and salary in descending order.
76  • select department_id,salary
77  from employees
78  order by department_id asc, salary desc ;
79  -- LIMIT:Display the first 5 employees hired in the year 2018.
80  • select employee_name
81  from employees
82  where year (hire_date) = '2018'
83  order by employee_name asc

```



```

84  limit 5;
85  -- 3. Aggregate Functions: Calculate the sum of all salaries in the Finance department.
86  • select sum(e.salary)
87  from employees e
88  join departments d on e.department_id = d.department_id
89  where d.department_name = 'Finance';
90  -- Find the minimum age among all employees.
91  • select min(age) from employees;
92  -- GROUP BY: List the maximum salary for each location.
93  • select location_id, max(salary) as max_salary
94  from employees
95  group by location_id;
96  -- Calculate the average salary for each designation containing the word 'Analyst'.
97  • select designation, avg(salary) as avg_salary
98  from employees
99  where designation like '%Analyst%'
100 group by designation;
101 -- HAVING: Find departments with less than 3 employees.
102 • select department_id, count(*) as employee_count
103 from employees
104 group by department_id
105 having count(*) < 3;
106 -- Find locations with female employees whose average age is below 30.
107 • select location_id, avg(age) as avg_age
108 from employees
109 where gender = 'F'
110 group by location_id
111 having avg(age) < 30;
112 -- Inner Join: List employee names, their designations, and department names where employees are assigned to a department.
113 • select e.employee_name, e.designation, d.department_name

```

```
113 • select e.employee_name,e.designation,d.department_name
114 from employees e
115 join departments d on e.department_id;
116 -- Left Join:List all departments along with the total number of employees in each department, including departments with no employees.
117 • select d.department_name,count(e.employee_id) as total_employees
118 from departments d
119 left join employees e on d.department_id = e.department_id
120 group by d.department_name;
121 -- Right Join:Display all locations along with the names of employees assigned to each location.If no employees are assigned to a location, display NULL for employee name.
122 • select l.location_name,e.employee_name
123 from employees e
124 right join location l on e.location_id = l.location_id;
125
126
```